

Newton's Laws of Motion, solved

Step-by-step solutions to every practice problem in Chapter 3. Try each problem on your own before you read its solution.

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P3.1

The two third-law forces act on **different bodies**: one on the cart, one on the horse. To decide whether a body accelerates, add up only the forces acting **on that body**.

- **On the cart:** the horse's forward pull P , and rolling friction f_c backward. The cart accelerates forward if $P > f_c$.
- **On the horse:** the cart's backward pull P (third-law partner of the horse's pull), and the ground's forward friction F_g on its hooves. The hooves push backward on the ground, and the ground pushes forward on the horse. The horse accelerates forward if $F_g > P$.

Taken together, horse and cart accelerate because the ground pushes the horse forward harder than the friction forces hold the system back. The internal pair, P and $-P$, cancels only when horse and cart are treated as a *single* system.

P3.2

FORCE ON THE BOOK	AGENT	THIRD-LAW PARTNER	PARTNER ACTS ON
Weight mg , down	Earth (gravity)	Book's gravitational pull on the Earth, mg , up	Earth

FORCE ON THE BOOK	AGENT	THIRD-LAW PARTNER	PARTNER ACTS ON
Normal force N , up	Table (contact)	Book's push on the table, N , down	Table

The weight and the normal force are equal and opposite here, but **not** because of the third law. They act on the same body, and they balance only because the book has no acceleration (second law).

P3.3

At 400 km the gravitational field is about $g(R_E/(R_E + h))^2 = 9.8(6371/6771)^2 \approx 8.7 \text{ m/s}^2$, about 89% of its surface value. Gravity has certainly not disappeared.

The station and everything in it are in **free fall**: they all have the same acceleration, $g_{400} \approx 8.7 \text{ m/s}^2$, towards the Earth's centre. They also move sideways fast enough (about 7.7 km/s) that they keep missing the Earth. That sideways motion is what makes it an orbit (Chapter 8).

For an astronaut standing on a scale, $N = m(g_{\text{local}} + a)$ with $a = -g_{\text{local}}$, so $N = 0$. The **apparent weight** is zero: nothing needs to push on the astronaut to keep them moving with the station. “Weightless” means *zero apparent weight*, not zero gravity.

P3.4

Ideal pulley. The tension is the **same** on both sides: a massless, frictionless pulley cannot sustain a net force along the string. The tension is **not** equal to either weight. It lies between them, $m_2g < T < m_1g$, so that the heavy mass accelerates down ($m_1g > T$) and the light mass up ($T > m_2g$). See Example 3.3.

Massive pulley. If the pulley has a moment of inertia and the string grips it without slipping, the tensions **differ**. The difference $T_1 - T_2$ provides the torque that gives the pulley its angular acceleration. We treat this case in Chapter 7.

P3.5

FBD. Three forces act on the crate: the tension T up the slope, the weight mg , and the normal force N perpendicular to the slope. Use axes along the slope (up positive) and perpendicular to it.

Along the slope:

$$T - mg \sin 30^\circ = ma \Rightarrow T = m(g \sin 30^\circ + a) = 10(4.9 + 1.5) = 64 \text{ N}.$$

Perpendicular to the slope, there is no acceleration:

$$N = mg \cos 30^\circ = 10(9.8)(0.866) = 84.9 \text{ N}.$$

P3.6

FBD of the car: tension T up and weight Mg down. With up positive, $T - Mg = Ma$.

Maximum upward acceleration, with $T = 12\,000 \text{ N}$:

$$a_{\max} = \frac{T_{\max}}{M} - g = \frac{12\,000}{800} - 9.8 = 15.0 - 9.8 = 5.2 \text{ m/s}^2.$$

Maximum downward acceleration with a taut cable. The condition is $T = M(g + a) \geq 0$, so $a \geq -g$. The car can accelerate downward at up to $g = 9.8 \text{ m/s}^2$, which is free fall. A faster downward acceleration would need the cable to *push*, which a cable cannot do.

Evaluate. Real lift standards keep accelerations to about $1\text{--}1.5 \text{ m/s}^2$ for comfort, and require safety factors of 8–12 on cable strength.

P3.7

(a) **Whole system:** $a = F/(m_1 + m_2) = 30/6.0 = 5.0 \text{ m/s}^2$.

Rear block (4.0 kg) alone: only the string tension acts horizontally, so $T = m_{\text{rear}}a = 4.0(5.0) = 20 \text{ N}$.

(b) In general, $T = m_{\text{rear}} \frac{F}{m_1 + m_2} = \frac{4}{6}F$. Setting $T = 15 \text{ N}$:

$$F_{\max} = \frac{15 \times 6.0}{4.0} = 22.5 \text{ N}.$$

(c) Pulling the 4.0 kg block, the string now accelerates only the 2.0 kg block, so $T = \frac{2}{6}F$ and

$$F_{\max} = \frac{15 \times 6.0}{2.0} = 45 \text{ N}.$$

Lesson. A string's tension equals the force needed to accelerate whatever is *behind* it. To pull a train of loads gently, put the heaviest load at the front.

P3.8

Tension. The rope holds a 5.0 kg mass in equilibrium, so everywhere along the ideal rope

$$T = mg = 5.0(9.8) = 49 \text{ N.}$$

Forces on the foot pulley. Two rope segments pull on it, each with tension T , at 25° above and 25° below the horizontal (both pointing away from the foot).

- The vertical components, $+T \sin 25^\circ$ and $-T \sin 25^\circ$, cancel.
- The horizontal components add:

$$F_{\text{traction}} = 2T \cos 25^\circ = 2(49)(0.906) = 88.8 \text{ N.}$$

At 10° : $F = 2(49) \cos 10^\circ = 96.5 \text{ N.}$

Evaluate. The traction force increases towards $2T = 98 \text{ N}$ as the strands become parallel. The pulley roughly doubles the effect of the hanging weight, and the clinician adjusts the angle and the mass to set the force. A traction force of order 90 N is typical for keeping the ends of a fractured femur aligned against the pull of the thigh muscles.

P3.9

Use $N = m(g + a)$ with up positive, so $a = N/m - g$.

Start of trip:

$$a_1 = \frac{660}{60} - 9.8 = 11.0 - 9.8 = +1.2 \text{ m/s}^2 \text{ (upward).}$$

End of trip:

$$a_2 = \frac{520}{60} - 9.8 = 8.67 - 9.8 = -1.13 \text{ m/s}^2 \text{ (downward).}$$

Direction of travel. A scale reading measures only acceleration, not velocity. But we know the lift starts *from rest*. An upward acceleration from rest means it starts moving **up**. At the end, the downward acceleration slows that upward motion to a stop. So the lift is **going up**. A downward trip would show the low reading first.

P3.10

Ground frame. The bob accelerates with the train. The tension's horizontal component supplies ma , and its vertical component balances mg :

$$\tan \theta = \frac{a}{g} \Rightarrow a = g \tan 8.0^\circ = 9.8(0.1405) = 1.38 \text{ m/s}^2.$$

Direction. The thread hangs towards the *rear*, so the tension pulls the bob forward and up. The train is accelerating **forward**.

Braking. The acceleration points backward, so the thread swings **forward**, towards the front of the train. In the train's frame, the pseudo-force now points forward.

P3.11

Driving forces along the string. The component of each weight along its own face is:

- $m_1: m_1 g \sin 30^\circ = 5.0(9.8)(0.5) = 24.5 \text{ N}$
- $m_2: m_2 g \sin 60^\circ = 3.0(9.8)(0.866) = 25.46 \text{ N}$

The m_2 side wins narrowly, so m_2 slides down its face and m_1 moves up its face.

Equations, taking each block's direction of motion as positive:

$$m_2 : m_2 g \sin 60^\circ - T = m_2 a, \quad m_1 : T - m_1 g \sin 30^\circ = m_1 a.$$

Adding:

$$a = \frac{25.46 - 24.5}{8.0} = \frac{0.96}{8.0} = 0.12 \text{ m/s}^2.$$

$$T = m_1(g \sin 30^\circ + a) = 5.0(4.9 + 0.12) = 25.1 \text{ N}.$$

Check with m_2 : $25.46 - 25.1 = 0.36 \text{ N}$, and $m_2 a = 3.0(0.12) = 0.36 \text{ N}$. ✓

Evaluate. The lighter block wins because it sits on the steeper face. What matters along the string is $mg \sin \theta$, not mg .

P3.12

Pulley FBD. The pulley is ideal (massless). The string pulls down on it with T on each side, and the rod pulls up with F_{rod} . A massless pulley must have zero net force, so

$$F_{\text{rod}} = 2T = 2(36.75) = 73.5 \text{ N.}$$

Comparison. The total weight of the masses is $(5.0 + 3.0)(9.8) = 78.4 \text{ N}$, so the rod carries 4.9 N less than this.

Why. Treat the two masses together. Their centre of mass accelerates **downward**, because the heavier mass goes down:

$$a_{\text{cm}} = \frac{m_1 a - m_2 a}{m_1 + m_2} = \frac{(5.0 - 3.0)(2.45)}{8.0} = 0.6125 \text{ m/s}^2 \text{ down.}$$

Newton's second law for the pair gives

$$(m_1 + m_2)g - 2T = (m_1 + m_2)a_{\text{cm}} \Rightarrow 78.4 - 73.5 = 8.0(0.6125) = 4.9 \text{ N. } \checkmark$$

A support carries the full weight only if the centre of mass is not accelerating. This is the same physics as the lighter scale reading in a lift that is accelerating downward.

P3.13

Equilibrium of the junction. Let T_1 be the cable at 30° to the vertical and T_2 the cable at 45° , with $W = mg = 9800 \text{ N}$.

$$\text{Horizontal: } T_1 \sin 30^\circ = T_2 \sin 45^\circ \Rightarrow T_1 = 1.414 T_2.$$

$$\text{Vertical: } T_1 \cos 30^\circ + T_2 \cos 45^\circ = 9800.$$

Substitute the first into the second:

$$T_2 (1.414 \times 0.866 + 0.707) = 1.932 T_2 = 9800 \Rightarrow T_2 = 5070 \text{ N, } T_1 = 7170 \text{ N.}$$

Check with Lami's theorem. Each tension is proportional to the sine of the angle opposite it:

$$T_1 = W \frac{\sin 45^\circ}{\sin 75^\circ} = 9800 \frac{0.707}{0.966} = 7170 \text{ N. } \checkmark$$

Which cable? The cable closer to vertical (30°) carries more, about 73% of the weight compared with 52%. Note also that the two tensions add to **12 240 N**, well over the 9800 N load. Angled cables

always carry more than their “share”, and the effect grows rapidly as the cables flatten. Riggers keep sling angles steep for exactly this reason.

P3.14

Key observation. The block is at rest on a *frictionless* slope only because the whole system accelerates horizontally. The normal force must then supply both the vertical support and the horizontal acceleration.

Ground frame. The forces on the block are its weight mg (down) and the normal force N , perpendicular to the incline. N is tilted from the vertical by θ towards the direction of acceleration. With the block’s acceleration a horizontal:

$$\text{Vertical: } N \cos \theta = mg, \quad \text{Horizontal: } N \sin \theta = ma.$$

(a) Dividing the equations: $a = g \tan \theta$.

(b) $N = \frac{mg}{\cos \theta}$, which is *larger* than mg .

(c) The whole system (block plus wedge) has mass $M + m$ and acceleration a , with no friction from the floor:

$$F = (M + m) g \tan \theta.$$

Wedge frame. The frame accelerates at a , so add the pseudo-force ma , pointing opposite to a , to the block. In this frame the block is in equilibrium under mg , N and ma . Resolve along the incline: $mg \sin \theta = ma \cos \theta$, so $a = g \tan \theta$. Resolve perpendicular to it:

$$N = mg \cos \theta + ma \sin \theta = mg(\cos \theta + \sin \theta \tan \theta) = mg / \cos \theta. \quad \checkmark \text{ Same results.}$$

Evaluate. As $\theta \rightarrow 0$, $a \rightarrow 0$: a flat surface needs no acceleration. As $\theta \rightarrow 90^\circ$, $a \rightarrow \infty$: you cannot hold a block against a vertical frictionless wall by acceleration alone.

P3.15

Constraint. Let y_M be the depth of the movable pulley below the ceiling, and y_m the depth of m below the fixed pulley. The string has two segments supporting the movable pulley, so

$$L = 2y_M + y_m + \text{const} \Rightarrow \ddot{y}_m = -2\ddot{y}_M.$$

If m descends with acceleration a_m , then M rises with $a_M = a_m/2$.

Forces (one tension T throughout):

$$m \text{ (down positive)} : mg - T = m a_m = 2m a_M,$$

$$M \text{ (up positive)} : 2T - Mg = M a_M.$$

Solve. From the first equation, $T = mg - 2ma_M$. Substitute into the second:

$$2mg - 4ma_M - Mg = Ma_M \Rightarrow a_M = \frac{(2m - M)g}{4m + M}.$$

Numbers:

$$a_M = \frac{(6.0 - 4.0)(9.8)}{12 + 4.0} = 1.225 \text{ m/s}^2 \text{ up}, \quad a_m = 2.45 \text{ m/s}^2 \text{ down},$$

$$T = m(g - a_m) = 3.0(9.8 - 2.45) = 22.05 \text{ N} \approx 22 \text{ N}.$$

Check M : $2T - Mg = 44.1 - 39.2 = 4.9 \text{ N}$, and $Ma_M = 4.0(1.225) = 4.9 \text{ N}$. ✓

Evaluate. The system balances when $2m = M$. The smaller mass can lift the larger because of the pulley's 2:1 mechanical advantage.

P3.16

(a) Newton's second law, $m dv/dt = F_0 e^{-t/\tau}$, with $v(0) = 0$:

$$v(t) = \frac{F_0}{m} \int_0^t e^{-t'/\tau} dt' = \frac{F_0 \tau}{m} (1 - e^{-t/\tau}).$$

(b) As $t \rightarrow \infty$:

$$v_\infty = \frac{F_0 \tau}{m}.$$

Notice that $F_0 \tau = \int_0^\infty F dt$ is the total **impulse** delivered, so v_∞ is simply impulse $\div$ mass (Chapter 6).

(c) Integrate the velocity:

$$x(t) = \frac{F_0 \tau}{m} \left[t - \tau (1 - e^{-t/\tau}) \right].$$

At $t = \tau$:

$$x(\tau) = \frac{F_0 \tau}{m} [\tau - \tau + \tau e^{-1}] = \frac{F_0 \tau^2}{e m} \approx 0.368 \frac{F_0 \tau^2}{m}.$$

P3.17

(a) **On the floor.** The whole rope accelerates at $a = F/M$. Cut it at distance x from the pulled end. The section beyond the cut has length $L - x$ and mass $M(L - x)/L$, and only the tension at the cut pulls it forward:

$$T(x) = \frac{M(L - x)}{L} \cdot \frac{F}{M} = F \left(1 - \frac{x}{L}\right).$$

The tension falls linearly from F at the pulled end to zero at the free end.

(b) **Hanging.** The rope is in equilibrium. The tension at depth y supports the rope below that point, of length $L - y$:

$$T(y) = \frac{Mg(L - y)}{L}.$$

This is Mg at the ceiling and zero at the bottom.

Why the ideal-string rule fails. “Same tension everywhere” follows from applying $\sum F = ma$ to a small piece of string of *zero mass*. A piece of real rope has mass, so the tension must vary along it, either to accelerate each piece (a) or to support its weight (b). The ideal-string model is good whenever the rope’s mass is small compared with the masses it connects.

P3.18

Method. Work in the lift’s frame and add the pseudo-force mA downward to each mass ($A = 2.2 \text{ m/s}^2$ upward). Everything then behaves as if gravity were

$$g_{\text{eff}} = g + A = 9.8 + 2.2 = 12.0 \text{ m/s}^2.$$

Relative acceleration and tension. Use the Atwood results with g_{eff} :

$$a_{\text{rel}} = \frac{(m_1 - m_2) g_{\text{eff}}}{m_1 + m_2} = \frac{2.0(12.0)}{8.0} = 3.0 \text{ m/s}^2,$$

$$T = \frac{2m_1 m_2 g_{\text{eff}}}{m_1 + m_2} = \frac{2(5.0)(3.0)(12.0)}{8.0} = 45 \text{ N}.$$

Ground frame. Relative to the lift, m_1 accelerates down at 3.0 and m_2 up at 3.0. Adding the lift’s 2.2 upward:

$$a_1 = 2.2 - 3.0 = -0.8 \text{ m/s}^2 \text{ (0.8 m/s}^2 \text{ down)}, \quad a_2 = 2.2 + 3.0 = 5.2 \text{ m/s}^2 \text{ up}.$$

Check m_1 in the inertial frame: $m_1g - T = 49 - 45 = 4.0$ N, and $m_1|a_1| = 5.0(0.8) = 4.0$ N, downward. ✓

Evaluate. An upward-accelerating lift increases both the relative acceleration and the tension, exactly as if gravity were stronger. This is Einstein's equivalence principle in its simplest form: inside a closed box, uniform acceleration cannot be distinguished from a gravitational field.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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